

TuniCars+

Resilience Testing Report

Docker Swarm vs Kubernetes

Session ID

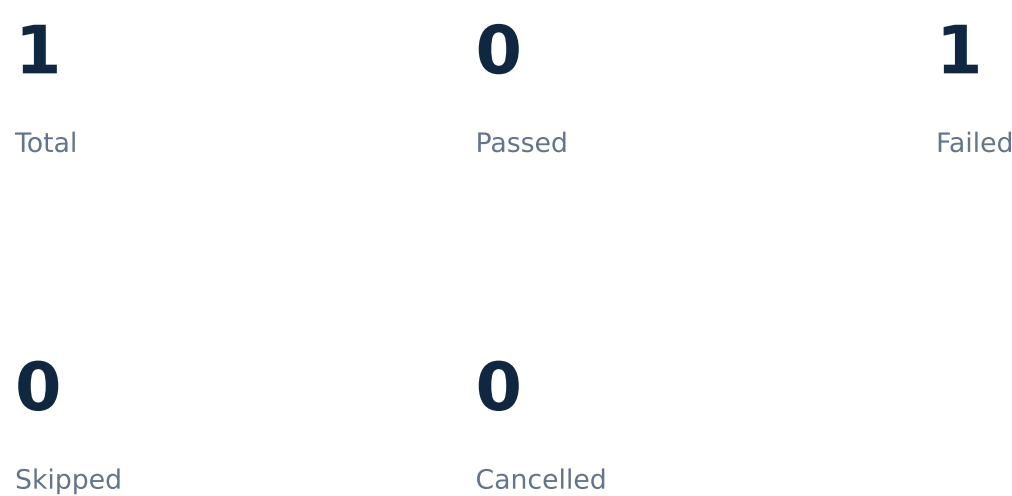
20260813_131722_d45382

Date

2026-08-13T12:17:22.943845+00:00

Executive Summary

20260813_131722_d45382



Overall measured availability: 98.57%

During this campaign, 0 of 1 scenarios passed, 1 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.

Test Environment

Architecture and technologies

Frontend → Backend → PostgreSQL

Docker Swarm environment

- Docker services
- FastAPI backend
- Next.js frontend
- PostgreSQL database

Kubernetes / Minikube environment

- Kubernetes Deployments and Services
- Minikube nodes
- FastAPI · Next.js · PostgreSQL

Test Scenarios

Scenario	kubernetes
memory	1

Full Results Table

Scenario	Platform	Detection	Recovery	Required	Success	Avg ms	P95 ms	HTTP Fail	Avail. %	Status	Skip Reason	Error Message
memory	kubernetes	N/A	N/A	N/A	N/A	107.112	35.66	1	98.57	FAIL	N/A	Stress pod finished with phase Running.

kubernetes — memory

State: **Fail**
Detection Time (s): N/A

Recovery Time (s): N/A

HTTP Success: N/A

HTTP Failures: 1

Availability (%): 98.57

CPU (%): N/A

Memory (MB): N/A

Recorded interpretation

Stress pod finished with phase Running.

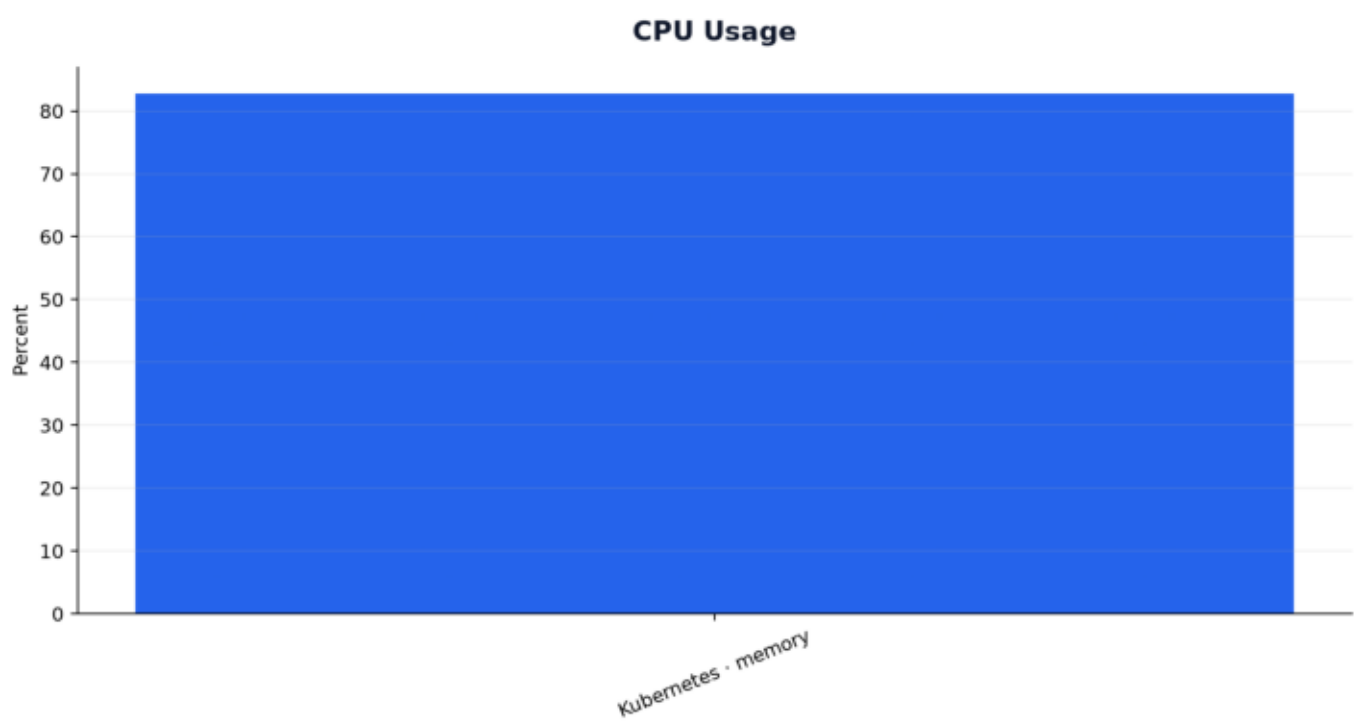
Availability

Current-session measured data



Cpu Usage

Current-session measured data



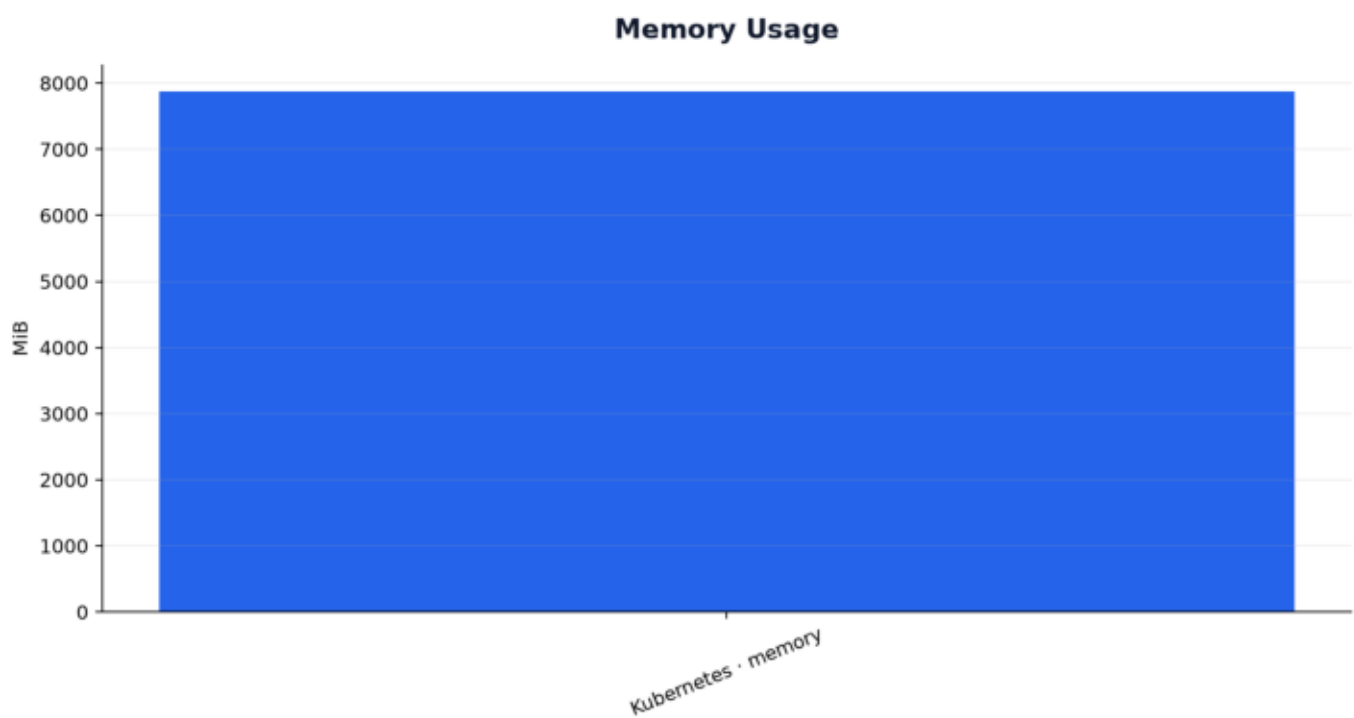
Http Failures

Current-session measured data



Memory Usage

Current-session measured data



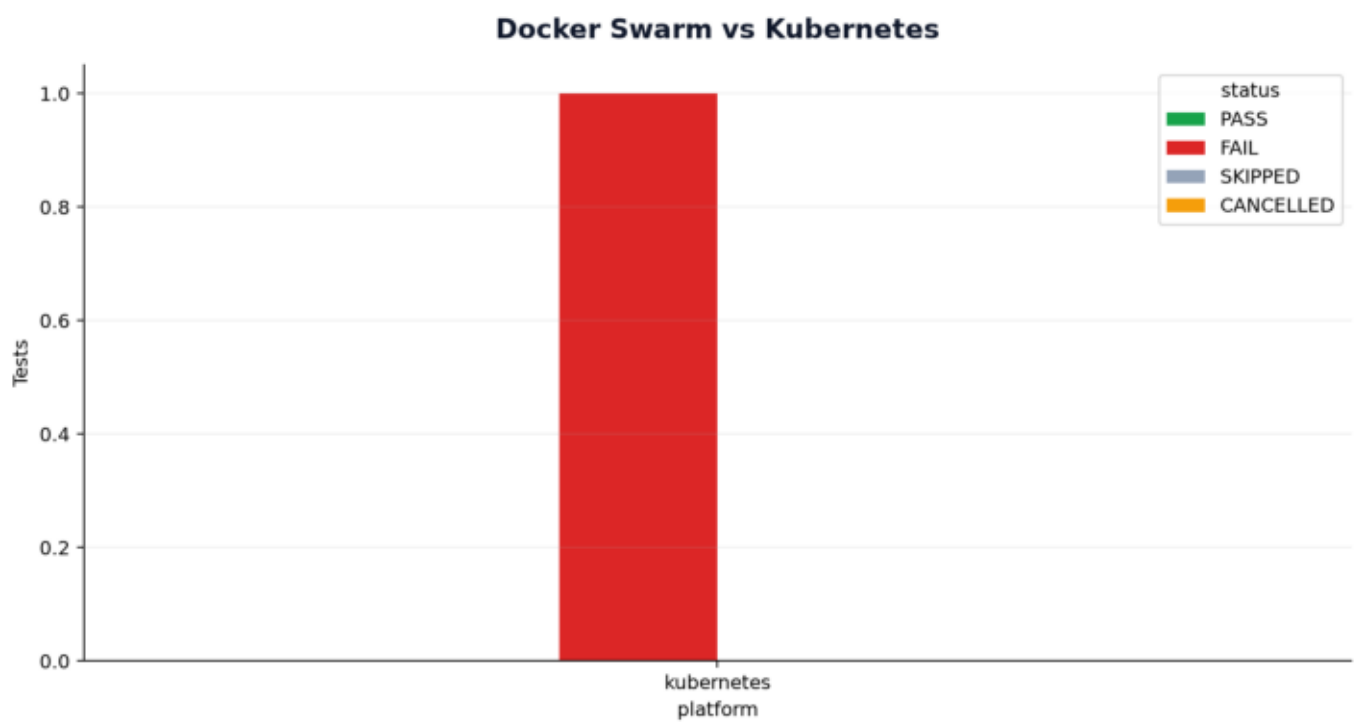
P95 Response Time

Current-session measured data



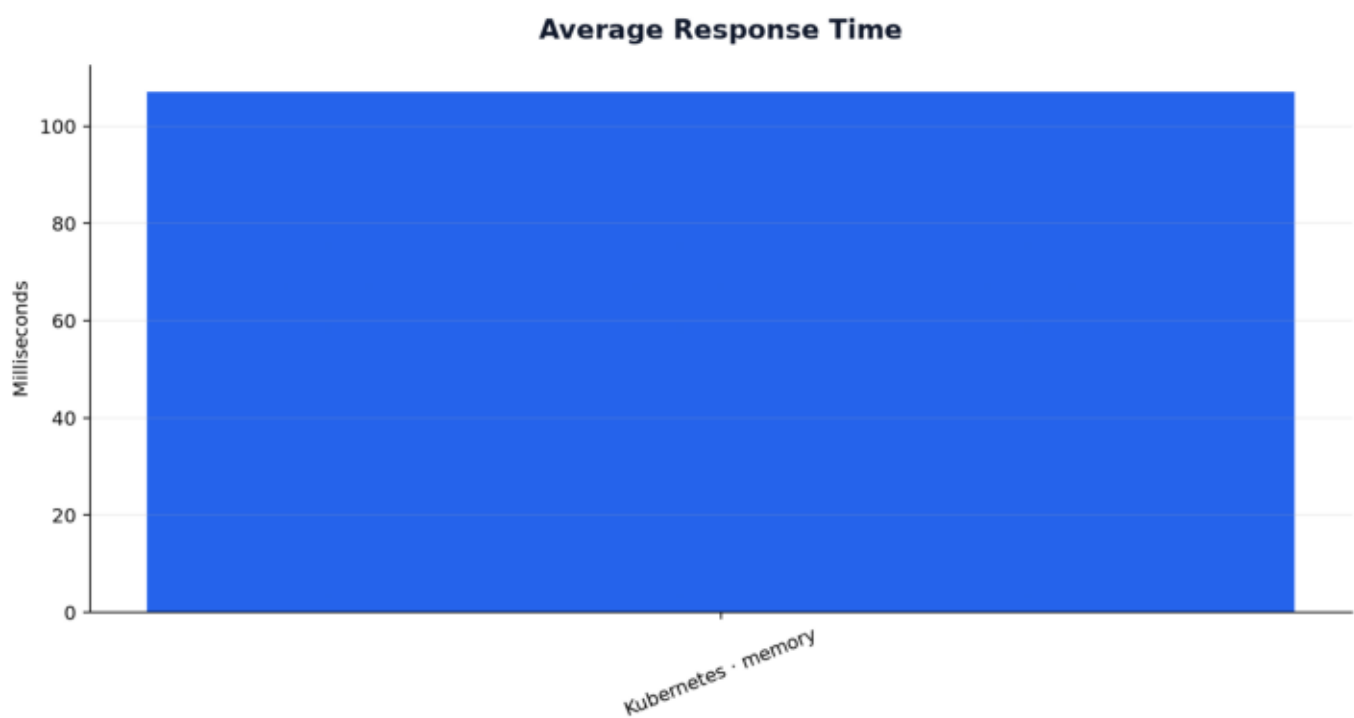
Platform Comparison

Current-session measured data



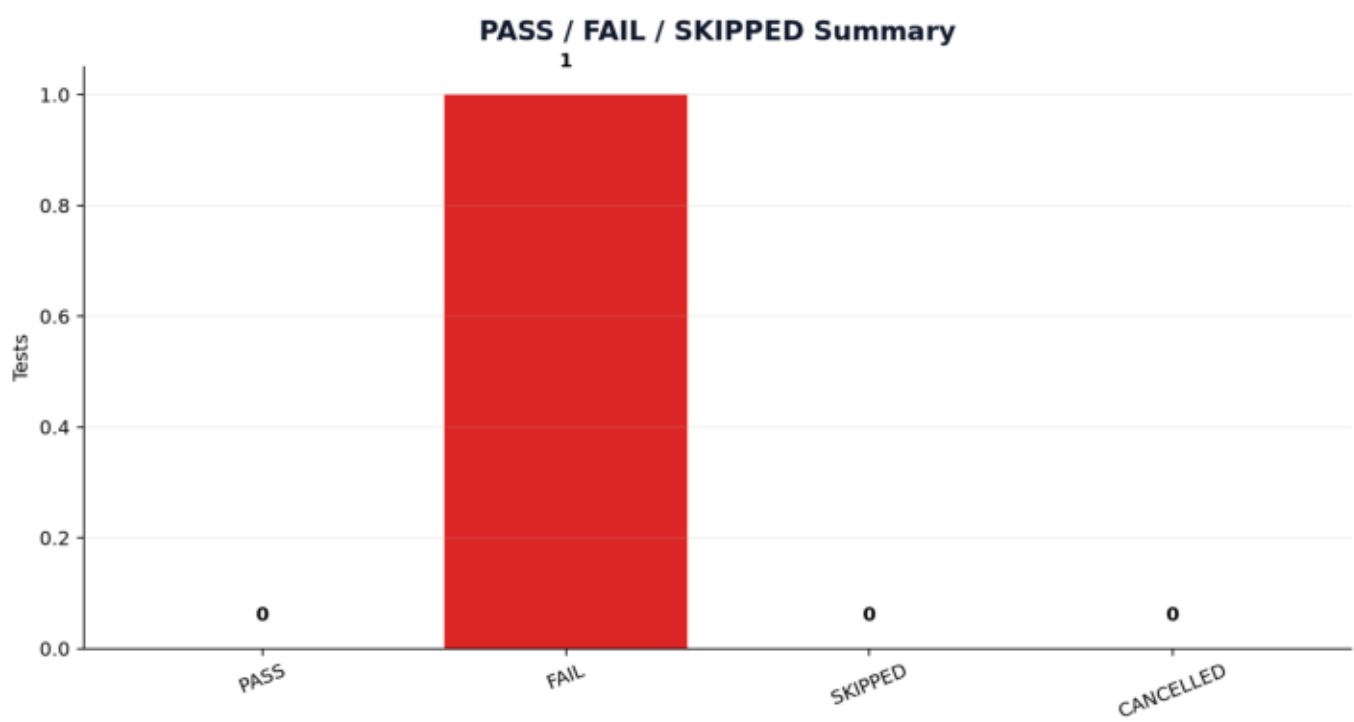
Response Time

Current-session measured data



Status Summary

Current-session measured data



Failures and Errors

Platform	Scenario	Status	Error
kubernetes	memory	FAIL	Stress pod finished with phase Runn

Skipped Tests

Platform	Scenario	Exact reason
—	—	No skipped tests recorded.

Swarm vs Kubernetes Comparison

Platform	FAIL
kubernetes	1

Final Conclusion

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During this campaign, 0 of 1 scenarios passed, 1 failed and 0 were skipped. Conclusions are limited to the measured scenarios and recorded environment constraints.